

Quantum Subgroup Meeting

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Discussion this week

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The Holevo Bound

Mutual Information: The Classical Case

Mutual information (MI) quantifies the amount of information obtained about one random variable by observing another. It measures the reduction in uncertainty.

- The **Shannon entropy** $H(X)$ measures the uncertainty of a random variable X :

$$H(X) = - \sum_x p(x) \log_2 p(x)$$

- For two random variables X and Y , the mutual information $I(X; Y)$ is defined as:

$$I(X; Y) = H(X) + H(Y) - H(X, Y) = H(X) - H(X|Y)$$

Where $H(X|Y)$ is the conditional entropy.

Quantum Mutual Information

Quantum mutual information is the generalization of classical mutual information to quantum systems. It measures the total correlations (classical and quantum) between two subsystems of a composite system.

- The **von Neumann entropy** $S(\rho)$ measures the uncertainty of a quantum state ρ :

$$S(\rho) = -\text{Tr}(\rho \log_2 \rho)$$

- For a bipartite state ρ_{AB} , the quantum mutual information $I(A; B)$ is defined as:

$$I(A; B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$$

The Holevo Bound

The Holevo bound, or Holevo's theorem, sets a limit on the amount of classical information that can be obtained by measuring an ensemble of quantum states. It provides an upper bound on the accessible information.

- The classical mutual information $I(X; Y)$ between the preparation of quantum states (X) and the measurement outcomes (Y) is bounded by the Holevo information $\chi(\mathcal{E})$:

$$I(X; Y) \leq \chi(\mathcal{E})$$

- The Holevo information is defined as:

$$\chi(\mathcal{E}) = S(\bar{\rho}) - \sum_x p_x S(\rho_x)$$

Where $\bar{\rho} = \sum_x p_x \rho_x$ is the average state of the ensemble, and ρ_x is the state prepared with probability p_x .

Example: Ensemble of 3 States

Consider a simple ensemble of three pure states, $\{\rho_1, \rho_1\}$, $\{\rho_2, \rho_2\}$, $\{\rho_3, \rho_3\}$, with equal probabilities $p_x = 1/3$. The states are arranged symmetrically on the equator of the Bloch sphere.

The three states are:

$$\rho_1 = |0\rangle \langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\rho_2 = \frac{1}{2}(|0\rangle + |1\rangle)(\langle 0| + \langle 1|) = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\rho_3 = \frac{1}{2}(|0\rangle - |1\rangle)(\langle 0| - \langle 1|) = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

Example: Calculating Holevo Bound (1/2)

The Holevo information is $\chi(\mathcal{E}) = S(\bar{\rho}) - \sum_x p_x S(\rho_x)$.

- Each state is pure, so its von Neumann entropy is zero:

$$S(\rho_x) = -\text{Tr}(\rho_x \log_2 \rho_x) = 0$$

- Thus, the second term of the Holevo information is zero:

$$\sum_x p_x S(\rho_x) = \frac{1}{3}S(\rho_1) + \frac{1}{3}S(\rho_2) + \frac{1}{3}S(\rho_3) = 0$$

Example: Calculating Holevo Bound (2/2)

- We calculate the average state $\bar{\rho} = \sum_x p_x \rho_x$:

$$\begin{aligned}\bar{\rho} &= \frac{1}{3} \left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \right) \\ &= \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} = \frac{1}{2} \mathbb{I}\end{aligned}$$

- The von Neumann entropy of the maximally mixed state $\bar{\rho}$ is:

$$S(\bar{\rho}) = -\text{Tr} \left(\frac{1}{2} \mathbb{I} \log_2 \frac{1}{2} \mathbb{I} \right) = -\text{Tr} \left(-\frac{1}{2} \mathbb{I} \right) = 1 \text{ bit}$$

- Therefore, the Holevo bound is:

$$I(X; Y) \leq \chi(\mathcal{E}) = S(\bar{\rho}) - 0 = 1 \text{ bit}$$

Discussion this week
My Conclusion

Summary of our work

- $(P_{\text{succ}})^{\text{MED}} = (P_{\text{succ}})^{\text{MED}+}$. See the proof in Chapter 5.
- $(P_{\text{succ}})^{\text{MED}} \geq (P_{\text{succ}})^{\text{OptUQSD}}$. See the proof in Chapter 5.
- Trade-off MED and UQSD
- The quantum circuit complexity is on par with circuits for isometries.

Trade-off MED and UQSD

- Here, P is the probability matrix for the POVM defined by the set of operators $\{\Pi_i\}$, and Q is the target UQSD probability matrix.
- $\|P - Q\|_1$ is the L_1 distance between P and Q .
- $1/w \leq a$ is OptUQSD, where a is usually 2.
- $1/w \geq b$ is MED, where the value b varies case by case.

$$\begin{array}{ll} \text{maximize} & P_{\text{succ}} - w\|P - Q\|_1 \\ \{\Pi\} & \end{array} \quad (1)$$

$$\begin{array}{ll} \text{subject to} & \Pi_i \succeq 0, \quad i = 1, \dots, k+1 \end{array} \quad (2)$$

$$\sum_{i=1}^{k+1} \Pi_i = I \quad (3)$$

The relations of the parameters and the solutions

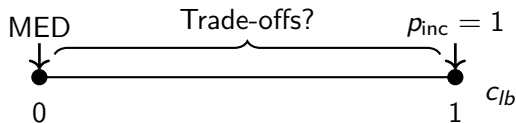


Figure: Fixed rate of inconclusive outcome (FRIO)

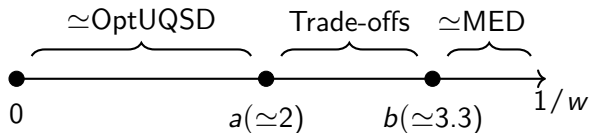


Figure: Our proposal to trade off MED and UQSD.

Meeting notes

- The definition of mutual information seems to be off or inappropriate.
- Add circuit examples to my thesis.